ESC103: Engineering Mathematics and Computation

Exam Review

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Fall 2026

Contents

1 Vectors **2**
- 1.1 Vector Operations . . . 2
  - 1.1.1 Vector Projection . . . 2
- 1.2 Vector Properties . . . 3
- 1.3 Plane Geometry . . . 3
  - 1.3.1 Interlude (Row and Column Picture) . . . 3

2 Matrices **4**
- 2.1 Matrix Operations . . . 4

1 Vectors

1.1 Vector Operations

Here are some basic vector operations in $\mathbb{R}^2$.

- Vector Addition: $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}$
- Scalar Multiplication: $c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$

Both of these operations are commutative and associative.

There are two ways to multiply vectors in $\mathbb{R}^3$. The dot product takes two vectors and returns a scalar. Here are two definitions for the dot product.

Definition. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$. The algebraic definition of the dot product is

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3.$$

Let θ be the angle between $\mathbf{v}$ and $\mathbf{w}$. The geometric definition of the dot product is

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \|\mathbf{w}\| \cos\theta.$$

The dot product is commutative and distributive over vector addition.

The cross product takes two vectors and returns a vector that is orthogonal to both. Here are two definitions for the cross product.

Definition. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$. The algebraic definition of the cross product is

$$\mathbf{v} \times \mathbf{w} = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix}.$$

Let θ be the angle between $\mathbf{v}$ and $\mathbf{w}$. The geometric definition of the cross product is

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \|\mathbf{w}\| \sin\theta.$$

The cross product is anti-commutative and not associative.

1.1.1 Vector Projection

The normalization of a vector, $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

The scalar projection of $\mathbf{v}$ onto $\mathbf{w}$ is

$$\text{comp}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{w}\|} \text{ (algebraic)} \quad \text{comp}_{\mathbf{w}}\mathbf{v} = \|\mathbf{v}\| \cos\theta \text{ (geometric)}.$$

Combining both, the formula for the vector projection of $\mathbf{v}$ onto $\mathbf{w}$ is

$$\text{proj}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{w}^2\|}\mathbf{w}.$$

1.2 Vector Properties

Here are some special types of vectors in $\mathbb{R}^3$.

- Zero Vector: A vector with length 0, denoted by $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
- Unit Vector: A vector with length 1, denoted by $\hat{\mathbf{v}}$
- Standard Basis Vectors: Special unit vectors denoted by $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

The length of a vector $\mathbf{v}$ in $\mathbb{R}^3$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

Two vectors $\mathbf{v}$ and $\mathbf{w}$ are orthogonal if

$$\mathbf{v} \cdot \mathbf{w} = 0.$$

Two vectors $\mathbf{v}$ and $\mathbf{w}$ are collinear if

$$\mathbf{v} \times \mathbf{w} = \mathbf{0}.$$

1.3 Plane Geometry

A line in $\mathbb{R}^3$ can be represented by the vector equation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + a\mathbf{d} \quad a \in \mathbb{R},$$

where $\mathbf{d}$ is a direction vector and (x_0, y_0, z_0) is a point on the line.

Similarly, a plane in $\mathbb{R}^3$ can be represented by the vector equation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + a\mathbf{d}_1 + b\mathbf{d}_2 \quad a, \in \mathbb{R},$$

where $\mathbf{d}_1$, $\mathbf{d}_2$ are non-collinear direction vectors and (x_0, y_0, z_0) is a point on the plane.

A plane in $\mathbb{R}^3$ can also be represented by the scalar equation

$$ax + by + cz = d,$$

where $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ is a normal vector of the plane (why?).

1.3.1 Interlude (Row and Column Picture)

The row picture of a system of linear equations treats each equation as a geometric line or plane, and the solution is the point where they all intersect.

The column picture of a system of linear equations views the system as a linear combination of column vectors, and to solve you need to find the correct scalar multiple of each vector.

2 Matrices

2.1 Matrix Operations

Here are some notes on matrices.